

REVIEW

A review to the methods of personality measurements in wild animals

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Abstract

One of the challenges in the study of animal personalities is to select a suitable method to measure personality. Here, we reviewed the three common methods (traits rating, behavioral coding, and experimentation) used in animal personality measurement in wild animals. For each method, we summarized its experimental animals and conditions, advantages, and disadvantages in personality measurement. One method could be used to measure the personalities of different animal species, and different methods could be used to measure the personality of one animal species. Overall, traits rating was used to test the personalities of large animals (e.g., primates), behavioral coding was conducted both in large and small animals (e.g., primates, birds, rodents, and fishes), whereas experimentation was used in small animals (e.g., rodents, birds, amphibians, and insects). Although the methods of experimentation were originally designed for model animals (e.g., mice and rats), some of them were used to test the personalities of wild animals under indoor conditions, in which the experimental animals were transferred from the field to the laboratory. Unfortunately, few studies have estimated the applicability of these methods to wild animals. In the future, effective methods of personality measurement are needed for wild animals, especially the methods that can be used in the field.

KEYWORDS

animal personality, behavioral coding, experimentation, personality measurement, traits rating

INTRODUCTION

Consistent differences of the behavior of individuals within a population under varied environments or at different times are referred to as animal personality (Gosling, 2001; Wolf & Weissing, 2012). Generally, a broad personality dimension includes all behaviors for which animals have stable individual differences, while a narrow personality dimension indicates a limited number of personality axes, such as boldness, activity, exploration, aggressiveness, or sociability (Carter et al., 2013; Gosling, 2001; Réale et al., 2007, 2010). Although narrow personality definitions are not exhaustive, they have been widely adopted in both animal personality and animal behavior studies (Carter et al., 2013). For a given animal, boldness is the responses to dangerous situations in a familiar environment, activity indicates the level of

individual activity in a non-novel and risk-free environment, exploration reflects the tendency to explore new environments or the responses to novelty, aggressiveness stands for the intensity of responses to conspecifics, and sociality indicates responses to the presence of conspecifics (excluding aggressive behavior) (Réale et al., 2007).

A growing body of research has shown that personality is closely related to foraging strategy, reproduction, heredity, development, population dynamics and distribution, ecological function, and evolution of wild animals (reviewed in Cheng et al., 2023; Dochtermann et al., 2015; Moiron et al., 2020; Sih et al., 2004; Wolf & Weissing, 2012). For example, Ballew et al. (2017) found that bold individuals of largemouth bass (*Micropterus salmoides*) had a higher chance to reproductive success than shy ones, and consequently individual boldness was highly

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consistent within the population and heritable across generations. In rodents, compared with shy individuals, bold individuals usually consume more seeds when they encounter them but hoard more seeds in places (e.g., open fields, forest gaps, and shrub edges) far from mother trees (Brehm & Mortelliti, 2022; Hunter et al., 2022; Zwolak & Sih, 2020). In highly disturbed habitats, shy and nonaggressive animals are more likely to survive because they are more cautious when exploring new environments, while bold and aggressive animals have more chance in population recovery due to their high rates of exploration and reproduction (de Azevedo & Young, 2021; MacKinlay & Shaw, 2022). Therefore, it can be advantageous to consider the personalities of wild animals when studying ecological processes or animal conservation.

A challenge in animal personality studies is how to measure the personalities of each individual accurately (Cheng et al., 2023; MacKinlay & Shaw, 2022). Traits rating, behavioral coding, and experimentation are the three most common methods employed in animal personality measurement (Blaszczyk, 2020; Carter et al., 2013; Gosling, 2001). Traits rating uses ratings of multiple items to describe the dimensions encompassing multiple personalities of animals (Carter et al., 2013; Vazire et al., 2007). This method should be conducted by observers who are familiar with the tested animals. Behavioral coding is conducted by recording the behavior of a specific individual according to a predetermined ethogram (Gosling, 2001; Vazire et al., 2007). Whereas, the method of experimentation is conducted under a controlled experimental condition to assess variability in a limited number of personality characteristics, such as boldness, activity, and exploration (Carter et al., 2013; Sih et al., 2004).

The methods of traits rating and behavioral coding have been summarized extensively in previous studies (e.g., Blaszczyk, 2020; Gosling, 2001; Uher & Asendorpf, 2008), while from the literature on experimentation it is unclear if the method is suitable for wild animals. In this study, we first outlined traits rating, behavioral coding, and experimentation, and then summarized the use of each method in different animal taxa, advantages, and disadvantages for the study of wild animals. We also make some suggestions for personality measurement in wild animals and hope to provide a reference source for researchers in related studies.

METHODS USED IN ANIMAL PERSONALITY MEASUREMENT

Traits rating

Traits rating requires observers to distinguish individuals based on a set of defined traits. A rating scale is used in the traits rating. Each personality descriptive item is rated on a seven-point scale with seven defined as “displays extremely large amounts of the trait” and one defined as “displays either total absence or negligible amounts of the trait” (Ebenau et al., 2020; Garai et al., 2016). It has been used to measure

Practitioner points

- We outlined three personality measurement methods: traits rating is used in large animals, behavioral coding is used in different animal taxa, and experimentation is widely used in small animals.
- Researchers could select the appropriate personality measurement methods to study the effects of animal personality on populations, communities, and ecosystems based on the description in this paper.
- Develop new methods and techniques for personality research in wild animals by using new computer technology and data processing technology in the future.

activity, sociability, and/or dominance of primates (e.g., *Semnopithecus entellus*, *Cebus capucinus*, and *Pan paniscus*) and domestic animals (e.g., cats and dogs) (Garai et al., 2016; Gosling, 2001; ěKonečná et al., 2008; Manson & Perry, 2013) (Figure 1). For example, Garai et al. (2016) selected five from 31 factors of the items from hominoid personality questionnaire and three factors from observed behaviors, and found that primates (e.g., *P. paniscus*) irritability was lower and introversion was higher in males than females, friendliness was lower and aggressiveness was higher in adults than adolescents, and social status rank was associated with aggressiveness among males. This method is efficient because it allows observers to draw on their experiences of animals over time and across contexts (Gosling, 2001). However, some studies argued that traits rating highly depends on observer experiences and subjective impressions, thus, the results will be more reliable after aggregating observations from multiple observers (Gosling, 2001; Vazire et al., 2007).

Behavioral coding

Behavioral coding records frequency and duration of a behavior by encoding the ethogram and performing a series of tests on animals (Ebenau et al., 2020; Gosling, 2001). It has been used to measure boldness, activity, aggressiveness, sociability, and/or anxiety of primates (e.g., *Macaca nigra*), rodents (e.g., *Mus domesticus*) and fishes (e.g., *Gasterosteus aculeatus*) (Benus & Røndigs, 1997; Huntingford & Giles, 1987; Neumann et al., 2013) (Figure 1). For example, Pritchard et al. (2014) found that self-directed behaviors, proximity, chasing, fear-grinning, touching, and lunging can predict personality scores of primates (e.g., *Macaca thibetana*). Behavioral coding simplifies a stream of actions and situations into discrete behavioral categories that can be counted, but inevitably misses the nuance of specific behaviors. Thus, it is not easy to choose the suitable behaviors for coding in studies of wild animals.

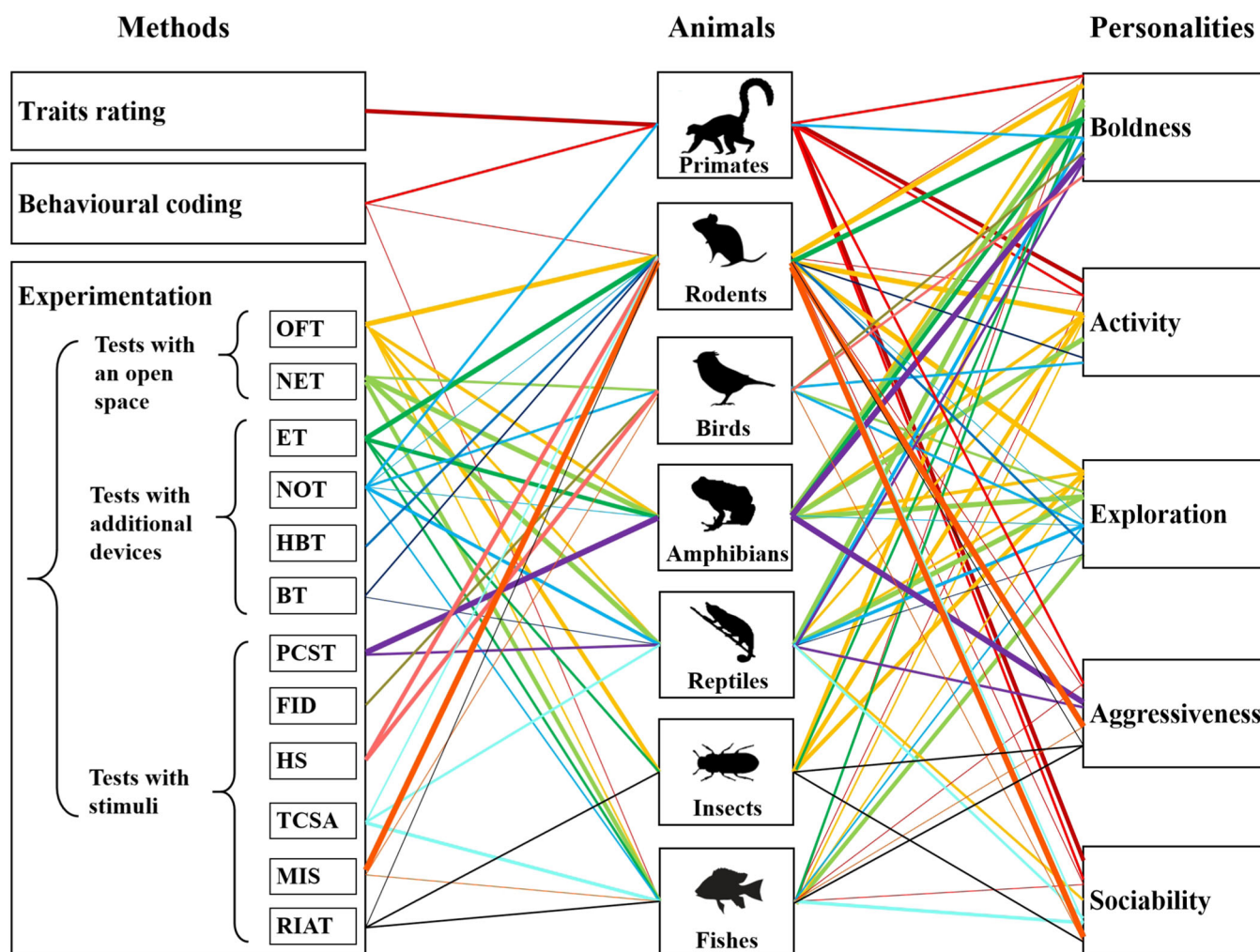


FIGURE 1 A conceptual framework of the relationships between the methods of personality measurements and the personality of different animals. The same color lines indicate the different personality measurements of different animal taxa by the same method. The different thickness of lines indicates the extent to which different species are studied. (The cartoons of animal taxa cited from Antonelli et al., 2022; Matsumura et al., 2017; Wan et al., 2022.) BT, barrier test; ET, emergence test; FID, flight initiation distance; HBT, hole-board test; HS, handling stress; MIS, mirror-image stimulation test; NET, novel environment test; NOT, novel object test; OFT, open-field test; PCST, physical and chemical stimuli test; RIAT, resident-intruder aggression test; TCSCA, three-chamber social approach.

Experimentation

Experimentation is designed to test the specific personality of animals under experimental conditions. It has been used to measure boldness, activity, exploration, aggressiveness, and sociability of rodents (e.g., *Peromyscus maniculatus*), birds (e.g., *Poecile gambeli*), reptiles (e.g., *Chrysemys picta*), and fishes (e.g., *Gambusia affinis*) (Brehm & Mortelliti, 2022; Cote et al., 2010; Fox et al., 2009; Roth et al., 2020) (Figure 1). Personalities of one animal species can be measured using different experimental methods, or the same method can be applied to measure different personalities. For example, the exploration in birds can be measured by their responses to novel objects and/or new environments (Groothuis & Carere, 2005; Klueen & Brommer, 2013). Both activity and exploration of rodents can be measured using the open-field test (Brehm & Mortelliti, 2022; Budaev, 1997; Maes et al., 2013). According to specific animals and personalities, logical experimental schemes and devices should be designed when conducting

experimentations. In addition, experimentation usually needs to transfer wild animals from the field to experimental devices, and this may result in errors because of the influences of investigators and experimental conditions on the animals. Thus, acclimation to experimental devices is necessary when measuring the personalities of wild animals under indoor conditions.

The difference of the three methods

In general, the traits rating was used to measure the personality of large animals (e.g., primates) without artificial manipulation or additional experimental devices. Traits rating requires observers to be familiar with the test animals and their behaviors that are associated with the target personalities. Moreover, the traits rating is inefficient and there is a need to assess whether the ratings are reliable and valid (Gosling, 2001).

Behavioral coding has been used in different taxa, such as primates, rodents, and fishes. It is often considered as a

more objective assessment technique than traits rating. However, behavioral coding is relatively inefficient and it is challenging to select suitable behaviors to reflect the tested personality of animals (Carter et al., 2013; Gosling, 2001).

Experimentation is widely conducted in rodents, birds, amphibians, and other small animals. We can accurately and quickly measure a certain personality of these animals under experimental conditions. However, it is necessary to select suitable methods of experimentation according to different personalities and animals. For example, the higher number of times of rear rate (the rate an individual rears during test duration) indicates both higher activity and enhanced exploration of the rodent in the open-field test (Boone et al., 2022; Brehm et al., 2019). We use the mean speed (dividing the total distance traveled in the test by test duration) of an open-field test to reflect the activity of individuals, but use the head-dip rate (dividing the number of times the animal head enters the hole by test duration) in hole-board test to reflect rodent exploratory behavior. It is also necessary to select experimental devices that are suitable for animal taxa under given experimental conditions. For example, when using open-field tests to measure animal personality, the size of the experimental field is different for insects and rodents that differ in their mobility. At the same time, for some reptiles with poor vision but developed sense of smell, choosing chemical stimuli is more effective than physical stimuli.

Challenges of personality measurements in wild animals

The common problems for many methods of personality measurement in wild animals are the effects of stress, noise, and non-target influences (Réale et al., 2007). Under controlled conditions, animals perform differently from those under the normal conditions because they are vulnerable to high stress during the period of testing. One behavior could indicate different personalities in different animal species and individuals. During the daytime, for example, increased grooming indicates low anxiety in nocturnal deer mice (*P. maniculatus*), but high anxiety in diurnal voles (*Myodes gapperi*) (Brehm et al., 2019). To reduce the effects of environmental noise, experimental conditions should be consistent and quiet. Sometimes, animals which do not behave as we expected do not mean they have not got the personality traits we defined. In a barrier test, for example, individuals which frequently cross the barrier may indicate high activity, but individuals who do not cross the barrier but move frequently may also have high activity (Herde & Eccard, 2013). To improve the accuracy of animal personality measurement, appropriate behavioral indicators, repeated measurements using different methods (e.g., behavioral coding and experimentation), and repeated measurements conducted by different observers

should be considered. In addition, all tests of wild animals should follow the international guides of wild animal ethics.

METHODS OF EXPERIMENTATION

We highlight experimentation hereafter because the methods of traits rating and behavioral coding have been well reviewed by Gosling (2001). To better understand how to measure the personality of animals with different experimental devices, we divided the method of experimentation into three categories of tests with an open space, tests with additional devices, and tests with stimuli. For each of these methods, we outline its manipulation steps, targeted animal taxa and personalities, and advantages and disadvantages.

Tests with an open space

Tests with an open space means putting an experimental animal in an open field to test its behavioral performances related to a specific personality, including open-field tests and novel environment tests.

Open-field test

The open-field test is one of the common methods of measuring boldness, activity, and exploration of wild animals (Figure 1). The animal for testing is placed in an open field for acclimation for a period of time (e.g., 2–5 min) and then its behaviors related to a given personality are recorded/quantified. Animal boldness is quantified by the proportion of time in the center and latency time to cross a center line (Brehm et al., 2019; Walsh & Cummins, 1976), and animals which spend more time in the central area of the test arena are higher in boldness (Boone et al., 2022; Mella et al., 2016; Schirmer et al., 2019). The open-field test has been used to test the boldness of rodents (e.g., *P. maniculatus*), amphibians (e.g., *Bufo calamita*), insects (e.g., *Eriopsis connexa*), and fishes (e.g., *Poecilia reticulata*) (Brehm et al., 2019; Budaev, 1997; Maes et al., 2013; Rodrigues et al., 2016). Materials of the testing instrument may affect the measurement of animal boldness. For example, transparent walls of the testing arena are likely to expose the experimental animals to high predation risk, which could alter its behavioral performances in boldness.

Animal activity is scored by mean speed (total distance traveled during the test divided by test time), rear rate (the number of times forelegs left arena floor divided by test time), and total activity (total time of activity) (Boone et al., 2022; Wilson & Krause, 2012). The open-field test has been used to test the activity of several animal species, such as rodents (e.g., *Tamias sibiricus*), amphibians (e.g., *Rana ridibunda*), reptiles (e.g., *Lampropholis delicata*), insects (e.g., *Drosophila melanogaster* and *Tribolium castaneum*), and fishes (e.g., *P. reticulata* and *G. affinis*) (Anderson & Lawler, 2016; Budaev, 1997; Chen et al., 2018; Matsumura et al., 2017;

Michelangeli et al., 2020; Montiglio et al., 2012; Wilson & Krause, 2012).

Animal exploration is estimated using rear rate, sniff/chew (sniff floor or walls/chew or scratch at floor or walls), and number of grid squares explored of the tested animals in the open-field test, whereby the higher of these parameters indicates the higher exploration the tested animals have (Boone et al., 2022; Rodrigues et al., 2016). Animal exploration has been tested in rodents (e.g., *Melomys cervinipes* and *Myotomys unisulcatus*), amphibians (e.g., *B. calamita*), reptiles (e.g., *L. delicata*), insects (e.g., *Diploptera punctata*), and fishes (e.g., *P. reticulata*) using the open-field test (Agnani et al., 2020; Budaev, 1997; Maes et al., 2013; Michelangeli et al., 2020; Rowell & Rymer, 2021; Stanley et al., 2017).

Animals are required to acclimate to experimental conditions to reduce the impacts of new environments on behavioral performances before conducting open-field tests (Carter et al., 2013; Réale et al., 2007). The behavioral characteristics of the tested species should be carefully selected when designing open-field tests (Brodin, 2009; Carlson & Langkilde, 2013). For example, some researchers did not detect individual differences in boldness on bullfrog larva (*Rana catesbeiana*), possibly because bullfrog larva would never experience open-field test conditions (Carlson & Langkilde, 2013). This study quantified boldness by quantifying the behavior of bullfrog larva in the center of the open sites, but central areas were less risky for this species than marginal areas. Additionally, open-field test devices should be selected according to individual body-size and range of motion of the test animals (Carlson & Langkilde, 2013; Walsh & Cummins, 1976).

Novel environment test

The novel environment test is usually performed to measure exploration, boldness, and activity of wild animals by putting some specific decorations in the open field to simulate a novel environmental stimulus. The experimental animals are usually measured repeatedly (c.a., 3–5 times). The decorations of the test field are renewed to ensure the tested animal is in new environments in each test (Fox et al., 2009; Siviter et al., 2017). The novel environment test has been used to test the exploration of birds (e.g., *Parus major*), amphibians (e.g., *R. catesbeiana*), reptiles (e.g., *Phrynosoma cornutum*), and fishes (e.g., *P. reticulata*) (Barrett et al., 2022; Both et al., 2005; Brown & Irving, 2014; Carlson & Langkilde, 2013). For example, an individual of *P. major* was placed in a closed chamber with five artificial trees to judge its exploratory ability by recording the number of movements (jumps between branches in the trees), two types of behavior (e.g., flights and hops), and time required to visit the five trees within 10 min (Both et al., 2005; Groothuis & Carere, 2005). Usually, small animals (e.g., amphibians, *Allobates femoralis* and *Rana temporaria*; reptiles, *P. cornutum* and *Zootoca vivipara*) can be placed into a rectangular container or a strange cage space to observe and record individual behavior variables (Barrett et al., 2022; Brodin et al., 2013;

Kelleher et al., 2018; Le Galliard et al., 2013). Novel environment test is also used to measure the boldness of animals. For example, an individual's latency to resume activity or move time in a new environment is measured to indicate the boldness of amphibians (e.g., *R. ridibunda*) and reptiles (e.g., *C. picta* and *Pogona vitticeps*) (Roth et al., 2020; Siviter et al., 2017; Wilson & Krause, 2012). Several other studies used a novel environment test to measure the activity of amphibians (e.g., *Lithobates sylvaticus* and *R. catesbeiana*) and fishes (e.g., *G. affinis*) (Cote et al., 2010; Koprivnikar et al., 2012; Smith & Doupnik, 2005). However, when measuring activity in unfamiliar environments, it is not fully clear whether an individual's rate of movement reflects its baseline activity level or exploratory behavior stimulated by the new environments (Kelleher et al., 2018). Activity of animals may be measured more accurately in a familiar environment than in a new environment (Barrett et al., 2022; Carlson & Langkilde, 2013). Moreover, in each repeated test, how to change the decorations of the test field is controversial and its effects on the behavioral plasticity of the tested animals have not been fully estimated.

Tests with additional devices

Tests with additional experimental devices, such as the emergence test, novel object test, hole-board test, and barrier test, measure the personalities of animals using particular devices. Behavioral responses of the tested animals to the devices are recorded to reflect their personalities (e.g., boldness, activity, and exploration).

Emergence test

The emergence test is used to measure the boldness and anxiety of wild animals by recording latency to emerge from a shelter (Carter et al., 2013) (Figure 1). A tested animal is put into a shelter to measure latency to emerge from the shelter and time staying at the end of the shelter (without exiting). Individuals which have shorter latency to emerge from the tunnel and shorter time staying at the end of the shelter are higher in boldness (Brehm et al., 2019; Carter et al., 2013). The emergence test has been used to test boldness of rodents (e.g., *P. maniculatus* and *M. gapperi*), amphibians (e.g., *Rhinella marina* and *R. sphenoccephala*), fishes (e.g., *G. aculeatus* and *G. affinis*), and insects (e.g., *Pyrrhocoris apterus*) (Bell, 2005; Brehm et al., 2019; Chen et al., 2018; Davenport et al., 2020; Gruber et al., 2017; Gyuris et al., 2012). When testing insects, a subject was placed in a bottle covered with paper. After a minute of acclimation, the paper was lifted and appearance time of the insect emerging from the shelter was recorded in 10 min (Gyuris et al., 2012). If an individual develops a fear of external environments and refuses to emerge from the shelter, emerging time from the shelter should be discarded (Gyuris et al., 2012).

A special case of the emergence test is the dark-light test, which tests the time an animal moves from a dark

area to a light area to measure the anxiety of rodents (e.g., *Microtus arvalis*) (Gracceva et al., 2014). In that, a rodent is placed in a dark box to measure the latency of emerging from the dark box to the light areas (Gracceva et al., 2014). The tested animal is anxious if it has longer latency to emerge from the dark area (Gracceva et al., 2014). The dark-light test was also used to measure the boldness of amphibians and insects (Müller & Müller, 2015; Peignier et al., 2022). For example, Müller and Müller (2015) put an insect (*Phaedon cochleariae*) into a brown glass vial for 1 min, shook the vial quickly and placed it horizontally to record the latency of *P. cochleariae* to leave the vial completely. The subject is bolder if it has shorter latency to emerge from the vial. Overall, the dark-light test is usually used in small animals (e.g., insects and small rodents) that are less active. Besides, preference for dark or light might have different meanings of personality in diurnal and nocturnal animals.

Novel object test

The novel object test was used to test the responses of an experimental animal to new objects (e.g., foods and toys) under laboratory conditions. This test is typically performed to measure the exploration and boldness of wild animals (Figure 1). New foods are often brightly colored or dyed with food dyes, such as colored eggs, bread rolls, apples, and orange squash (Carter et al., 2012). New toys are plastic grasshopper, toy scorpion, ping-pong ball, Lego piece, clothes peg, and rubber lizard (Błaszczuk, 2020; Silva et al., 2020). Approach latency to novel objects and time of sniffing/touching novel objects were used to determine the personality of the tested animal. The novel object test has been used to measure personalities in primates (e.g., *Mungotictis decemlineata*), rodents (e.g., *M. cervinipes*), birds (e.g., *P. major*), amphibians (e.g., *R. marina*), reptiles (e.g., *L. delicata*), and fishes (e.g., *G. aculeatus*) (Groothuis & Carere, 2005; Gruber et al., 2017; Michelangeli et al., 2020; Nomakuchi et al., 2009; Rasolofoniaina et al., 2021; Rowell & Rymer, 2021). To avoid interference of the new environments, individuals need to acclimate to the test conditions for a period of time (e.g., 2–5 min) before conducting a novel object test. One of the challenges of this method is to judge whether an object is really strange to an animal because there are no standards for the shape, color, and size of a new object (Carter et al., 2013; Fox et al., 2009). It is also hard to distinguish between exploration and boldness of animals when they approach to the novel objects (Gruber et al., 2017; Siviter et al., 2017).

Hole-board test

The hole-board test is often used to measure exploration of rodents (e.g., *T. striatus*, *Rattus rattus*, and *M. arvalis*) by recording head-dip rate and head-dip time proportion for a tested individual by using a plastic plate with holes within an experimental chamber (Herde & Eccard, 2013;

Martin & Réale, 2008; Vobrúbová et al., 2021) (Figure 1). Individuals that have a higher rate or proportion of head-dip in the holes are higher in exploration (Herde & Eccard, 2013; Martin & Réale, 2008). Besides, latency to find one hole, latency to find all holes, and number of nose-in-hole events were also used to analyze the exploration of *M. arvalis* (Herde & Eccard, 2013). Diameter of the holes should be smaller than the body size of the experimental animals to prevent them from entering the hole directly, and is about 4-cm depth to provide enough exploratory space (Martin & Réale, 2008). However, long-term testing may let experimental animals habituate to the hole-board and the testing conditions, leading to less exploration by the subjects (Martin & Réale, 2008; Taylor et al., 2012).

Barrier test

The barrier test requires a low partition in the middle of an experimental chamber to act as a barrier for an experimental animal to cross, and has been used to measure activity and exploration of wild animals. According to a pseudo-random schedule, an animal is placed in one of the compartments, and is recorded until it jumps across the barrier to the other compartment. The barrier test has been used to measure the activity of rodents (Figure 1). For example, Herde and Eccard (2013) and Gracceva et al. (2014) tested the activity of *M. arvalis* by recording the latency to jump and the number of barrier crossings. Shorter latency to jump and higher number of crossings indicate higher activity. The barrier test was also used to measure the exploration of reptiles (e.g., *L. delicata*), in that the shorter the latency period of crossing the obstacle, the more exploratory the tested animal is (Chung et al., 2017). Behavioral indices of the barrier test are usually recorded using a stopwatch, but per-stopwatch reaction time of different experimenters may affect results. This shortcoming could be improved by conducting remote video monitoring.

Tests with stimuli

The tests with stimuli record behavioral responses of the subjects to special stimuli that are associated with threat and stress under captive conditions. Stimuli or stress could be generated by physical materials, chemicals, investigators, and social stimuli.

Physical and chemical stimuli

Physical and chemical stimuli are widely used in stress, aggressiveness, and boldness studies of wild animals (Kelleher et al., 2018) (Figure 1). External threats are used to stimulate individuals, and then individual behavioral responses to the stimuli are observed and recorded. Physical threat stimuli to experimental animals can be generated by modified metal rods to simulate predator attacks or by playing the sounds of territorial

invaders (Chaloupka et al., 2022; Kelleher et al., 2018). For example, in reptiles, Le Balle et al. (2021) measured the stress of *Chelonoidis carbonarius*, Roth et al. (2020) measured the aggressiveness of *C. picta* by slightly touching animals' noses with a tool. And in amphibians, Chaloupka et al. (2022) measured the aggressiveness of *A. femoralis* by broadcasting a standardized synthetic advertisement. Chemical threat stimuli to subjects can be generated by using the odor of predators and observing their behavioral responses (Langkilde & Carlson, 2014). For example, Langkilde and Carlson (2014) measured the boldness of amphibians (e.g., *L. sylvaticus*) using chemical threat stimulus by collecting water in which larval dragonflies (*Anax junius*) and adult eastern red-spotted newts (*Notophthalmus viridescens*) as a predation cue. One of the advantages of the threat stimulation test is that investigators can consistently provide similar simulated attacks, predator sounds, or chemical cues. However, it is worthy of consideration that appropriate stimuli should be selected for each animal species. For example, predator sounds or chemicals may be better than simulated attacks for visually degraded animals.

Flight initiation distance

Flight initiation distance to a threat has been used to test the boldness of wild animals by measuring their escape distance between the tested animal and the sources of the threat (e.g., an investigator or a predator) when an intruder is approaching the test animal (Blumstein, 2006; Bókony et al., 2012) (Figure 1). Flight initiation distance is usually used to test boldness in birds (e.g., *Junco hyemalis* and *Somateria mollissima*), whereby an individual which has a shorter flight initiation distance is higher in boldness (Atwell et al., 2012; Seltsmann et al., 2013). However, the flight initiation distance is often influenced by the ecological and social status of the tested animal, behavior of the investigator or predator, and environmental conditions. Another disadvantage of this method is the difficulty of getting repeated observations in wild animals, especially for those species living in groups. This method is more applicable to birds living in relatively closed habitats (e.g., in enclosures) or with low flight ability living in open fields.

Handling stress

The handling stress is often used to measure the boldness and docility of birds (Kluen & Brommer, 2013) (Figure 1). In which, a researcher holds a tested individual and counts the number of abdominal movements in 1 min. Animals are nervous when they are attacked by a predator or are held by an experimenter, and thus their breathing rate reflects their boldness at a certain level of predation risk (Edelaar et al., 2012). By artificially manipulating the stress of birds (e.g., *P. major*), Carere and Oers (2004) found that the individuals that have higher respiration rates are less bold. Most individual birds (e.g., *Tyto alba* and

Taeniopygia guttata) show changes in body temperature, elevated glucocorticoids, and accelerated respiratory rate in 3 min of being hand held, changing the frequency and strength of struggling to reflect their personalities in response to the applied stress (David et al., 2012; van den Brink et al., 2012). For example, individuals who struggle less are more docile in *Cyanistes caeruleus* (Brommer & Kluen, 2012; Class & Brommer, 2016). Behavioral changes are usually associated with physiological changes, for example, an accelerated respiratory rate leads to increased energy consumption. Thus, we can measure either behavioral responses (frequency of struggling) or physiological responses (elevated glucocorticoids) to measure the personality of animals. However, the handling stress test may cause damage to the animals, and it is difficult to control the amount of stress an animal is subjected to.

A special case of the handling stress is the handling bag test, which is designed to measure the docility of rodents under the conditions of human presence and manipulation (Brehm et al., 2019) (Figure 1). A handler suspends a bag containing an individual animal and measures the proportion of time that an individual remains stationary during 1 min; the more time of a motionless state reflects higher docility of the test animal (Brehm & Mortelliti, 2022; Martin & Réale, 2008). Handling bag tests have been used to measure docility in rodents (e.g., *T. striatus*, *Trichosurus vulpecula*, and *P. maniculatus*) (Brehm & Mortelliti, 2022; Martin & Réale, 2008; Mella et al., 2016). Tested animals are vulnerable to experimenters and environments because handling bags are often transparent, thus the animal should be acclimated to the tested conditions and environments before a handling trial.

Three-chamber social approach

As a classic experiment in animal behavior research, the three-chamber social approach is widely used to determine the sociability of wild animals (Leung et al., 2018; Teyssier et al., 2014) (Figure 1). A test arena is divided into three chambers of social, non-social, and neutral compartments. The social stimuli for the tested individual are unfamiliar conspecific individuals or a paper with the smell of the same-sex individual (Chung et al., 2017; Michelangeli et al., 2020; Teyssier et al., 2014). A tested individual is placed in the neutral compartment for acclimation for several minutes, then the barriers between the chambers are removed, and the tested individual freely selects between the social and non-social compartments. The tested individual is sociable if it spends more time in the social compartment. This method has been used in testing the sociability of several species, such as reptiles (e.g., *C. picta*, *L. delicata*, and *Z. vivipara*) (Chung et al., 2017; Roth et al., 2020; Teyssier et al., 2014). The stimulation in new environments may influence individual choice of social and non-social compartments (Carter et al., 2013; Teyssier et al., 2014). In addition, measurement of sociability by the three-chamber social approach is usually restricted to monitoring one or two variables (e.g., time spent in each

chamber and/or time spent in proximity to distinct wire cups), which may not be a very reliable reflection of social propensity (Jabarin et al., 2022). Sociability could be more accurately assessed by measurement of multiple behavioral variables. In group-living animals, the three-chamber social approach has also been used to measure preferences of the tested individual for group size and/or group characteristics (i.e., personality of the social group).

Mirror-image stimulation test

The mirror-image stimulation test is another method of presentation of social stimuli to measure the aggressiveness and sociability of wild animals, and recording rate and latency of attack of mirror images in an experimental arena. Individuals with higher attack rate or shorter attack latency are more aggressive (Boon et al., 2007; Svendsen & Armitage, 1973). The mirror-image stimulation test has been used to measure aggressiveness and sociability in rodents (e.g., *Tamiasciurus hudsonicus*, *Sciurus niger*, and *Dipodomys merriami*) and fishes (e.g., *Salmo salar*) (Church & Grant, 2018; Dochtermann et al., 2012; Hurtado & Mabry, 2017; Krause et al., 2015) (Figure 1). Sociability of animals was also estimated by measuring the percentage of time spent in front of the mirror and behind the mirror, and an individual has higher sociability when it spends more time in front of the mirror (Boon et al., 2007; Silva et al., 2020). Social behavior of the tested animals should be considered when conducting mirror-image stimulation tests because non-territorial and/or group-living animals (e.g., rodents such as *Tamias dorsalis*; birds such as *Estrilda astrild*) would be more sociable than territorial and/or solitary-living species (Funghi et al., 2015; Silva et al., 2020). One of the disadvantages of the mirror-image stimulation test is the unnatural response of the “other” individual and the almost inevitable unnatural positive feedback loop where the “opponent” invariably escalates when the focal individual escalates. Another query is the possibility that individuals may differ in whether they are fooled by the mirror. More socially sophisticated individuals might be better at realizing that the mirror image is not an opponent.

Resident–intruder aggression test

The resident–intruder aggression test has been used to test the aggressiveness of wild animals either in the field or under the laboratory conditions (Figure 1). Before testing, the test animal (i.e., native individual or resident) is allowed to move randomly in an experimental area for at least 10 min to establish its territory. During the period of testing, latency period and number of attacks on invaders (i.e., intruders such as competitors or researchers) are recorded to reflect the aggressiveness of the tested animal. A higher latency time or larger number of attacks indicates higher aggressiveness of the native individual (Keiser et al., 2018). The resident–intruder aggression test has been used to test the aggressiveness of rodents (e.g., *Rattus norvegicus*), insects (e.g., *Calopteryx splendens*), and fishes (e.g., *G. aculeatus*) (Carnevali et al., 2013; Keiser et al., 2018;

Lacasse & Aubin-Horth, 2012). For group-living animals, the resident–intruder aggression test measures sociability rather than aggressiveness of an individual (Stanley et al., 2017). When the resident–intruder aggression test is performed in the field, interferences of researchers and other animals, and influences of environments and weather conditions should be considered (Gołab et al., 2013).

Suggestions for experimentation

It is important to choose or develop methods of experimentation that are applicable for personality measurement in wild animals. Using one method to measure different personalities can improve the efficiency of experiments, but may result in errors or unrepeatability (Dingemanse & Wright, 2020). In the open-field test, for example, individual rearing scores in rats could indicate either high activity or high exploration (Brehm et al., 2019; Boone et al., 2022). Therefore, when measuring a given personality, multiple methods should be considered according to species and individuals (Carter et al., 2013). For example, boldness of animals can be tested by performing emergence test and open-field test, while exploration of animals can be tested with hole-board test and novel environment test. Personality of an animal is confirmed if it performs similarly under the conditions of different methods.

Environmental changes may affect the outcome of personality measurements in wild animals (Kelleher et al., 2018). In previous studies, animals were usually captured in the field and then moved to new places or into the laboratory for testing, but effects of these changes on animal personality have not been fully estimated (Carter et al., 2013; Le Balle et al., 2021). Mazza et al. (2020) demonstrated that individuals of *M. arvalis* altered their behaviors considerably and were consistently shyer and less exploratory in the laboratory than in the field. Controlled environmental variables or extending time of acclimation before test may reduce negative impacts of changes in environment. To reduce animal's fear of experimental environment, the inner wall of the arena can be decorated according to the field conditions or to simulate field environments by installing display screens. In addition, we can also use robots to control switch of experimental devices, and to shoot videos to reduce disturbance of researchers.

Manipulating the interval of repeated measurements may improve the accuracy of measurements. A short interval between two tests may cause individual habituation to the experimental conditions and increase interference (Taylor et al., 2012). A demonstration of animal personality is therefore most convincing when inter-test interval is relatively long compared with the lifespan of the species (Dingemanse & Wright, 2020). For rodents, 1-month apart could reduce the effects of the previous experiences on experimental animals (e.g., *P. maniculatus* and *M. gapperi*, Brehm et al., 2019).

To accurately measure the personalities of wild animals, it is necessary to continuously improve the methods of personality measurement. In the open-field test, for example, anxiety and fear of wild animals could be reduced by using a shelter or prolonging the time of acclimation to the experimental conditions when measuring the

exploration and activity of animals (Carter et al., 2013). In the emergence test, measurement of time emerging from a shelter should be discarded if the individual fears the external environments and refuses to come out of the shelter (Gyuris et al., 2012). In the threat stimulation test, the tested arena could be divided into low-, medium-, and high-risk areas according to distances away from the risk sources (e.g., predators, models, feces, or urine of predators), and animal personality traits (e.g., boldness and exploration) could be determined using latency time and frequency of visiting each area. Besides, most of the methods used to measure the personality of wild animals in the previous studies are originally designed for model animals (e.g., rats and mice), but few of them have been assessed whether they are appropriate for the tested animals and conditions. In the future, the applicability of the methods originally designed for model animals should be further estimated in wild animals and methods designed for wild animals are particularly necessary.

CONCLUSIONS AND PERSPECTIVES

We used the methods of traits rating, behavioral coding, and experimentation as a framework for this review of methods of personality measurements in wild animals, and summarized the methods of experimentation in detail. In summary, traits rating was used to measure the personality of large animals such as primates, behavioral coding was used in a variety of taxa, such as primates, rodents, and fishes, while experimentation was widely used in rodents, birds, amphibians, and other small animals.

Overall, it is rare to directly measure the personality of wild animals in the field. Although traits rating can directly observe the natural behavior to assess the personality of wild animals in the field, it is difficult to track and observe the behavior of wild animals for a variety of reasons, for instance, in animals with strong athletic ability, large range of activities, aggression at close range, or in dense habitats. Behavioral coding and experimentation need to transfer wild animals from the field conditions to indoor (or caged) conditions for personality testing, which might cause changes to animal personality, but the side effects are never estimated. In the future, the new techniques of computer and data processing will be used to measure animal personalities in the field. For example, artificial intelligence might be used to combine Global Positioning System (GPS) and unmanned vehicles to automatically monitor and track wild animals, and GPS combined with remote sensing technologies to automatically analyze the relationship between animal personality and environmental characteristics in the field.

AUTHOR CONTRIBUTIONS

Jiming Cheng: Investigation (equal); methodology (equal); visualization (equal); writing—original draft (lead); writing—review and editing (supporting). **Huimin He:** Writing—original draft (supporting). **Min Zhang:** Visualization (equal); writing—original draft (supporting); writing—

review and editing (supporting). **Baiqi Gan:** Writing—original draft (supporting). **Hongyu Niu:** Methodology (equal); project administration (equal); supervision (supporting); validation (equal); visualization (equal); writing—review and editing (supporting). **Hongmao Zhang:** Methodology (equal); project administration (equal); supervision (lead); validation (equal); visualization (equal); writing—review and editing (lead).

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable—no new data generated, or the article describes entirely theoretical research.

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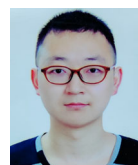
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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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